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Carbon-Neutral Hydraulic Fracturing: CO₂ Assisted Stimulation for Carbon Utilization and Sequestration

Abdul Muqtadir Khan, SLB, Sugar Land, Texas, USA

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Abstract

Carbon capture, utilization, and storage (CCUS) is a central technology pathway toward net-zero emissions, yet conventional deployment models, primarily involving deep saline aquifers or depleted reservoirs, face high screening, monitoring, and infrastructure costs. This paper proposes and evaluates a novel dual-purpose approach: replacing all hydraulic fracturing activity with CO₂ foam fracturing. CO₂ foam possesses superior stimulation properties, including reduced water demand, better leakoff control, faster cleanup, lower formation damage, and enhanced hydrocarbon recovery, while simultaneously enabling in-situ carbon storage. Modeling and literature evidence suggest that 40–50% of injected CO₂ could be residually trapped within stimulated formations due to hysteresis in relative permeability, micro-scale capillary entrapment, and adsorption mechanisms. Because candidate reservoirs for hydraulic fracturing already undergo rigorous barrier integrity and containment screening, many geological criteria for CO₂ storage are inherently met, reducing project risk and cost.

The paper develops a structured hypothesis supported by field analogues, laboratory studies, and integrated modeling workflows. Results indicate that when 40–50% of injected CO₂ remains permanently stored, the technique has the potential to render fracturing operations carbon-neutral or even carbon-negative. Global basin mapping shows significant scalability in regions with CO₂ sources near active unconventional development. A conceptual wellsite CO₂ capture system is also introduced to close the carbon loop with capturing, in addition to utilization and storage. Presented work reframes hydraulic fracturing as a strategic CCUS deployment pathway, offering production enhancement, water conservation, and material emissions reductions in a single operation. Field-scale pilots and policy incentives are recommended to accelerate adoption and fully quantify sequestration permanence.

Introduction

Hydrocarbon production faces growing pressure to curb its carbon footprint in line with global climate targets. Carbon Capture, Utilization and Storage (CCUS) has emerged as a pivotal strategy to achieve net-zero emissions, with the IPCC noting that meeting a 1.5°C warming limit is virtually impossible without CCUS contributions. The oil and gas sector, responsible for roughly 15% of global energy-related CO₂ emissions (Bello et al. 2024), is increasingly looking to CCUS to mitigate its impact while sustaining

energy supply. Notably, about 80% of CCUS projects worldwide already involve injecting captured CO₂ into oil and gas reservoirs for enhanced recovery (Hou et al. 2024), underscoring the synergy of carbon storage with petroleum operations. CCUS domain involves aspects related to capturing, effective utilization, and permanent storage of CO₂. CCUS has been a well-recognized and studied field within the petroleum industry. A typical application of effective utilization would be the use of CO₂ in an enhanced oil recovery project. Storage is done mostly in shallow sandstone water aquifers. Fig.1 shows the typical schematic demonstrating the value chain of typical CCUS operations highlighting the current, benchmark lifecycle of carbon under the lens of CCUS. Multiple aspects of capturing, transport, utilization and storage are detailed in the IEA report (2021).

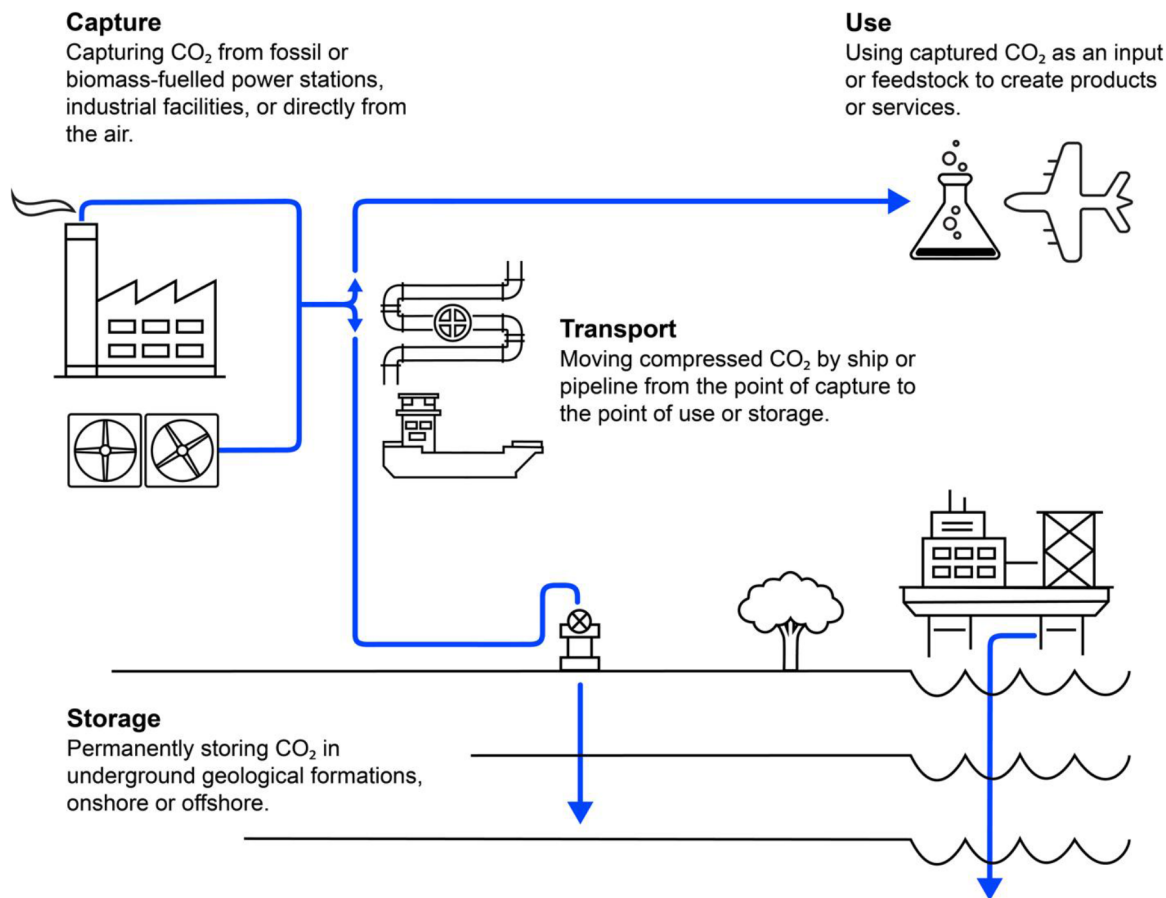


Figure 1—CCUS value chain depiction (IEA, 2021)

In this context, using CO₂ as a hydraulic fracturing fluid represents an integrative approach. It can utilize CO₂ for improving well productivity and concurrently sequester a substantial portion of that CO₂ in the subsurface. This paper hypothesizes that replacing conventional water-based fracturing fluids with CO₂ (particularly in foam form) offers dual benefits, boosting hydrocarbon production and trapping CO₂ underground, thus transforming fracturing from an emissions-intensive, water-intensive process into a more sustainable, carbon-managed technology. This paper is an attempt to present the idea that through hydraulic fracturing and stimulation, we can become carbon neutral or even carbon negative by utilizing the potential to reduce CO₂ flowback after fracturing. The underlying concept is to utilize and expand the potential of residual/hysteresis trapping property of CO₂ to decarbonize wellsite operations and reduce the CO₂ recovered after a fracturing, stimulation, or intervention treatments. A structured analysis of this concept is presented from theory to practical considerations.

CO₂ Fracturing and Stimulation for Carbon Utilization

In this section, we can elaborate how CO₂ fracturing satisfies the requirement for utilization, providing significant benefits from the production and formation damage perspective.

Benefits for Stimulation

Fracturing and stimulation with CO₂ foams has received significant attention from academic and industry circles because of its multiple advantages over N₂. A few of these are (a) higher hydrostatic pressure, (b) better expansion capacity during flowback, (c) higher interfacial tension reduction, (d) not requiring special high-pressure pumping equipment, and (e) pumping in liquid phase which allows achieving gas fractions that can create stable foams. The CO₂ foam fracturing technique is interesting to most operators as it provides many benefits in most conditions but has superior benefit in certain condition as detailed by [Khan et al. \(2023a\)](#). First, in mature depleted reservoirs where a post-stimulation nitrogen lift is required to initiate the fracture cleanup, the gas phase provides energy to the reservoir and enables the initiation of flowback. Second, where the rock is highly water wet and recovery of fracturing water is minimal leading to water blocks during the production phase, the CO₂, which has an extremely small surface tension and occupies the smaller/micro pores, enables better hydrocarbon and fracturing water recovery without damaging the reservoir. Third, in water-sensitive reservoirs with presence of high swelling-clay fractions, the low pH environment with CO₂ system minimizes the swelling of clays. Fourth, in gas condensate reservoirs experiencing condensate banking due to pressure dropping below dewpoint with depletion, liquid or supercritical CO₂ (scCO₂) is pumped and allowed to soak, interact, and become miscible with the condensate. The resulting mixture has a lower dewpoint, and condensate re-evaporates to be produced along with CO₂. It is important to note that this is a short-term solution because with passage of production period, the condensate block returns due to the higher dewpoint pressure of the current gas mixture. Also, the applicability of this method is when the banking is in the near-wellbore region only. Finally, in low-permeability gas reservoirs that require longer fractures to increase the contact area of the fracture with the reservoir, CO₂ foam provides better leakoff control (compared to the same viscosity regular crosslinked fluid) and enables generation of these long fractures. The reason a foam provides better leakoff control is because the fluid is compressible.

The first review and field implementation of foam fracturing was published by [Marsden et al. \(1967\)](#) but had limited utilization. [Gupta \(2003\)](#) shows that in 4 years starting in 1999, more than 350 treatments were executed to stimulate the Belly River and Bow Island formations. But this application was restricted for shallow wells (1000 m to 1200 m). Since then, in the 21st century, the adoption has increased as multiple basins are reaching maturity after decades of drawdown. North America had multiple treatments pumped, and applications have gained some momentum in the past decade in China, Central Asia, Western Siberia ([Oussoltsev et al. 2008](#)), and Middle East. [Rahim et al. \(2018\)](#) showed a comparison of acid fracturing cases in mature reservoir where CO₂ foams performed better. [Almutary et al. \(2021\)](#) performed a similar investigation but at a field scale with ~50 wells treated with CO₂ foams, N₂ energized fluid, and conventional acid fracturing. Their study focused on relevant aspects, in detail, such as production performance, fracture cleanup performance, flowback initiation efficiency, and freshwater utilization. The conclusion of both these studies was that CO₂ foams consistently performed better than N₂ energized fluid treatments generally. But this difference is significantly bigger in horizontal wells where the transit time of treating fluid is longer. The quality of CO₂ to preserve the energy better due to its supercritical nature at reservoir condition is demonstrated here. N₂ is in gas phase where the gas dissipates very quickly (especially in conventional reservoirs) during the operation of multiple stages across the lateral. Recently, CO₂ foam fracturing was also initiated in North Kuwait Jurassic gas formations as a solution to depleted reservoirs and was described as a significant technological advance ([Al-Muhanna et al. 2022](#)). There have not been any implementations of CO₂ fracturing in unconventional reservoirs, but extensive studies done by [Zhang et al. \(2017a\)](#) show

that supercritical CO₂ (scCO₂) has potential to enhance the induced fracture network due to potential phase transition to gaseous state.

CO₂ enhanced oil recovery (EOR) is one of the largest carbon utilization streams within the CCUS umbrella. With all the benefits provided through CO₂ fracturing, it could be easily treated as an analog of the EOR process. In CO₂ EOR, the field is developed with an injector: producer pattern where the CO₂ is injected in the injector wells, enhancing the mobility of oil, and pushing it toward the producer wells. When oil is produced through the producer wells, it also produces fractions of CO₂ along with it and so will be the case in CO₂ fracturing. The difference here being that the CO₂ will be injection into, and produced from, the same well. Both the processes are very similar and hence CO₂ fracturing and stimulation should not be left out of the utilization envelope.

CO₂ Fracturing for Sequestration

With all the executions and implementations till date, there have been many benefits that CO₂ foam has provided, and the method has unlocked the potential of reservoirs. But due to the physical properties and chemistry of CO₂, there are also some challenges associated with CO₂-assisted foams.

Carbon Sequestration Site Screening Criteria

Carbon capture and storage (CCS) projects typically have rigorous screening criteria to ensure injected CO₂ remains securely trapped underground for the long term. Asakura et al. (2023) details the key geologic criteria which include (a) a sufficiently deep reservoir (commonly >800–1000 m) so that CO₂ is in a dense supercritical state, (b) an adequate storage capacity and injectivity, and most critically (c) an impermeable caprock or sealing formation overlying the reservoir. The caprock integrity is usually evidenced by the reservoir's ability to retain hydrocarbons over geologic time. Additional favorable factors are (d) homogeneous geology to simplify CO₂ plume migration prediction, (e) low initial water saturation so CO₂ can occupy pore space, (f) presence of a supportive water aquifer to accommodate pressure, and (g) availability of existing wells and data as with depleted oil/gas fields. These conditions help ensure injected CO₂ does not leak in the upper stratigraphic layers or escape through undetected pathways.

Hydraulic fracturing candidate reservoirs inherently satisfy many of these criteria. By design, unconventional oil and gas targets (like tight shale or sandstone formations) are deep, low-permeability zones bounded by strong confining layers. In fact, fracture design explicitly relies on geologic barriers above and below the target zone to contain fracture growth (Khan et al. 2021). A strong, competent barrier is imperative to allow a long fracture generation to enhance the productivity from fracturing. This implies a screened fracturing candidate already comes with a "built-in" caprock or seal that can also serve to confine CO₂. Moreover, if the formation has trapped hydrocarbons (e.g. a tight oil reservoir), that is evidence of seal integrity similar to CCS screening (Askarova et al. 2023). Fracturing candidates are typically significantly deep (often 2–3 km for conventional sandstone and carbonate formations as well as unconventional shale gas/oil), satisfying the depth criterion for supercritical CO₂ storage. Largely, they also tend to be low in water content. For instance, shale gas has adsorbed gas instead of water, which aligns with the preference for low water saturation to maximize CO₂ storage capacity. Importantly, well construction and integrity for fracturing meet high standards with carbon steel casing and good cement isolation to prevent fluid migration, paralleling requirements for CO₂ injection wells. In summary, many generic CO₂ storage site requirements – such as depth, confining caprock, and containment away from groundwater, are inherently addressed when selecting a formation for hydraulic fracturing. This reduces the additional site screening burden if CO₂ is used as the fracturing fluid, since fracturing targets are pre-vetted to have competent seals and geologic containment. Overall, the overlap in screening criteria means that a fracturing site can double

as a CO₂ storage site with relatively few additional concerns, as long as careful monitoring ensures the induced fractures remain within the sealed zone.

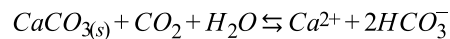
CO₂ Trapping Mechanisms

The primary objective of a CO₂ storage/sequestration project is to permanently keep the CO₂ subsurface. As any fluid CO₂ will have a tendency to flow, migrate and hence there are four mechanisms that enable the storage aspect of CO₂. One or all four mechanisms can be active at a given point wherein one mechanism tends to be dominant.

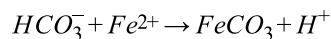
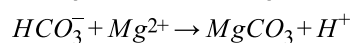
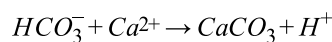
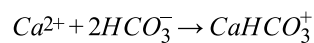
The first mechanism is the structural or stratigraphic trapping where a geologic layer such as a cap rock/anticline, fault, unconformity, pinch-out exists above the layer of storage. These geologic features have a low permeability and high capillary pressure which prevents the plume from migration permanently.

The second mechanism is the residual or hysteresis trapping. Here as the CO₂ continues to move, small portions of the CO₂ can be left behind in the disconnected pre networks, or as residual droplets in the pore spaces. Since this CO₂ is stored in smaller pores, and certain residual saturations they become immobile and stay back trapped in the rock. This is one of the most reliable mechanisms to permanently store the CO₂.

The third mechanism is the solubility trapping. In this process, the interfacial interaction of CO₂ and formation water is critical. At the interface, the chemical process takes over and CO₂ molecules dissolve in the formation water. In carbonate rock, weak carbonic acid (H₂CO₃) is formed as a result of this reaction which eventually turns into bicarbonate (HCO₃⁻). So here the CO₂ concentration is reduced due to conversion to the reaction products. Since it is a reversible process, some of the bicarbonate can also be converted back to CO₂ which is an essential consideration in this mechanism.



The fourth mechanism is the mineral trapping. Here the CO₂ eventually mineralizes to carbonate rocks such as calcite, dolomite, ferrous carbonate leaving it permanently stored. The bicarbonate ion reacts with the various divalent ions in the reservoir such as calcium, magnesium, iron etc. to form minerals.



Kelemen et al. (2020) presented a compiled schematic from review of different works (Fig. 4) and shows that in sedimentary rocks, different trapping mechanisms contribute from day1, but mineralization could take ~20-30 years to begin. Even though the mechanisms start from first day, to dissolve all the CO₂ in the water, it takes 1000s of years. In contrast, in a volcanic basalt rock the CO₂ reaction rate is very high and mineralization starts from day and complete mineralization is possible within a few years. This could be an important consideration for fracturing in geothermal basement rocks.

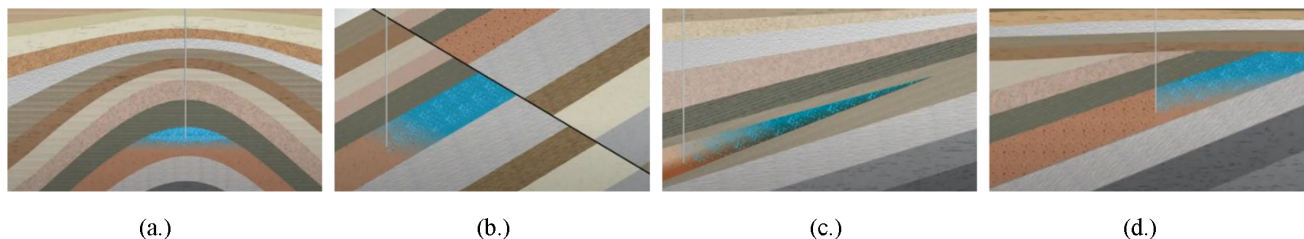


Figure 2—CO₂ structural or stratigraphic trapping features where the blue cloud represents the CO₂ plume: (a) Caprock forming an anticline. (b) Fault. (c) Pinch-out. (d) unconformity (Courtesy CO2CRC Ltd.).

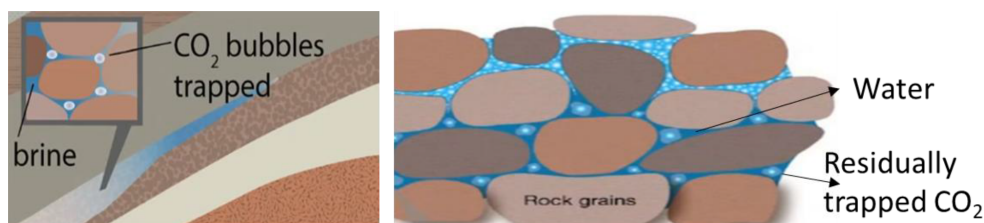


Figure 3—Residual trapping of CO₂ depicting the bubbles occupying small pore spaces in the intergranular rock spaces (Courtesy CO2CRC Ltd.).

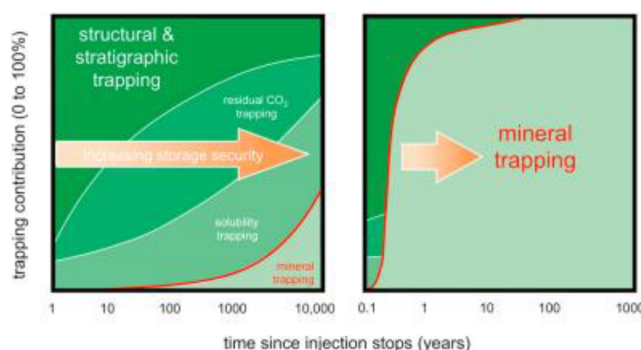


Figure 4—Trapping mechanisms contribution: (left) scCO₂ is injected in a typical sedimentary reservoir. (right) CO₂ in aqueous fluids in a peridotite or glassy basalt reservoir. (Kelemen et al. 2020)

The reason for long dissolution time is some volume of CO₂ being trapped in pores surrounded by water and have a faster dissolution due to the low IFT. But some CO₂ goes below the cap rock, and it is in contact with the water/brine (Fig. 5). When this CO₂ dissolves, water becomes denser and moves down in fingering pattern.

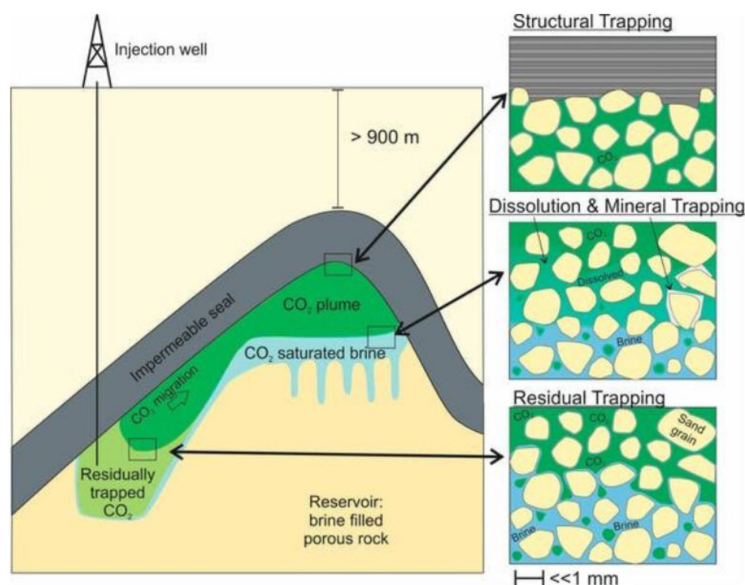


Figure 5—Trapping mechanisms contribution: (left) scCO₂ is injected in a typical sedimentary reservoir. (right) CO₂ in aqueous fluids in a peridotite or glassy basalt reservoir. (Flude 2020)

CO₂ Flow Behavior

An important consideration around the carbon storage discussions is the CO₂ wettability because it controls the capillary behavior of injected CO₂, including capillary entry pressure, relative permeability, and residual

CO₂ saturation. Contact angle measurements are the basic methodology that aids this understanding and can be upscaled to relative permeability-saturation curves. Alternately, elaborate core testing can also be conducted to generate these curves which can then be utilized for reservoir scale simulation and forward modeling for subsurface plume migration analysis. Two of the above mentioned mechanisms, i.e., structural, and residual trapping are related to the wettability. Capillary pressure (P_c) is an important dependency for these trapping processes. High P_c is required in the host rock (where CO₂ is injected) so that residual trapping will be better where CO₂ will be surrounded by water and will be easier to dissolve. Also, high P_c is required for the structural trap so that it will prevent CO₂ leakage. Most hydraulic fracturing targets low-permeability formations (e.g., shale and tight sandstones), which are characterized by elevated capillary entry pressures. This property inherently promotes CO₂ residual trapping, since high capillary resistance impedes its migration and enhances immobilization within pore spaces (Krevor et al. 2015)

In the equations below, σ is the interfacial tension (IFT) between CO₂ and water; θ is the contact angle; r is the pore radius; C_a is the capillary number; μ is the dynamic viscosity and V is the characteristic velocity. From the capillary number equation, knowing that the viscosity of CO₂ is extremely small (~ 0.04 cP for scCO₂ at 100° C and 3,250 psi and ~ 1 cp for 80% CO₂ foam at same conditions), capillary forces become dominant in dictating the flow inside the porous media. Since the capillary forces are a critical function of wettability (contact angle), it is an important consideration. Based on the below equation it can be observed that when θ is $\geq 90^\circ$, the P_c goes to 0 and hence this is a condition of CO₂ leak, i.e., it can be easily displaced by water or other reservoir fluids from the pores.

$$P_c = \frac{2\sigma \cos(\theta)}{r}$$

$$C_a = \frac{\text{Viscous forces}}{\text{Capillary forces}} = \frac{\mu \times V}{\sigma \times \cos(\theta)}$$

Another parameter controlling capillary action is the IFT and for perspective of CO₂ storage, sometimes it has a greater impact on P_c than the contact angle. A higher IFT is better for CO₂ storage since it will provide a higher capillary force. Fig. 6 below shows that at higher pressure, the CO₂ will be in supercritical phase and the density does not change considerably and hence the IFT stabilizes with increasing pressure. This region and resulting slope is of interest while dealing with scCO₂.

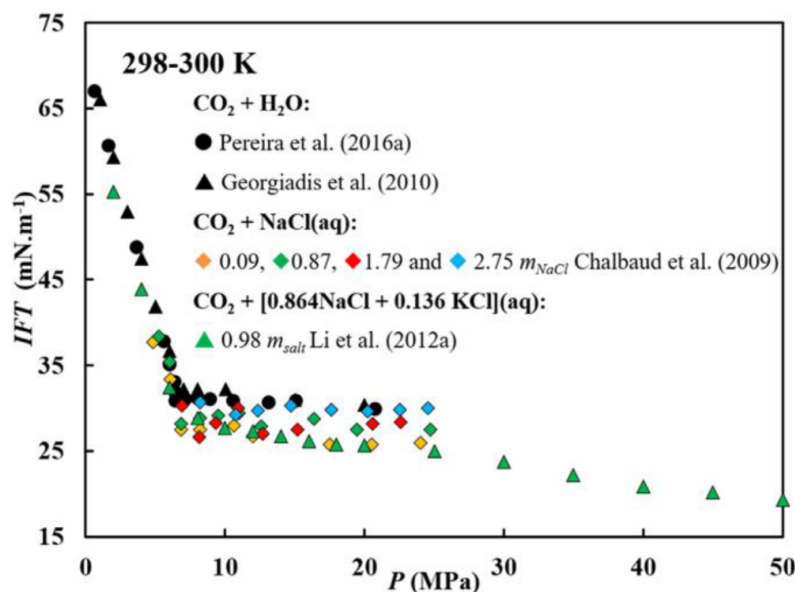


Figure 6—IFT-pressure diagram of CO₂ + H₂O and CO₂ + H₂O + salt systems were calculated considering the impact of CO₂ dissolution on the density of the aqueous phase. (Pereira et al., 2017).

Al-Menhali and Krevor (2016) did a study for carbonate oil reservoirs to show the impact of preferential rock wetting on the residual CO_2 saturations. Their work involved measuring the initial and residual saturations during a core flooding experiment. A water wet sample was used first without changing any surface chemistry of the rock. The next phase involved altering the rock surface to mixed wet. Fig. 7 shows a summary of the results achieved. They found that mixed wet will have less residual CO_2 trapping which is intuitive based on the capillary forces and snap-off mechanism. The snap-off mechanism is a pore-scale capillary instability where a non-wetting fluid phase, displaced by a wetting fluid during imbibition, breaks into disconnected droplets or ganglia at a pore-throat constriction. This fragmentation occurs when the invading wetting film thickens and bridges across the pore, effectively pinching off the non-wetting fluid due to a pressure difference between the leading and trailing parts of the fluid interface. Snap-off leads to capillary trapping of the non-wetting phase. A mixed wet rock will experience lower snap-off and does not allow trapping the CO_2 in small pores. In summary, a water wet rock is preferred for CO_2 storage due to the high residual trapping it provides. One critique on this study was that since crude oil or some other chemicals were used during the surface chemistry alteration, the porosity changes after this treatment were not checked. So, if the crude oil was not completely removed the porosity could have been changed and the change in residual saturation could be fully or partially a function of this porosity change.

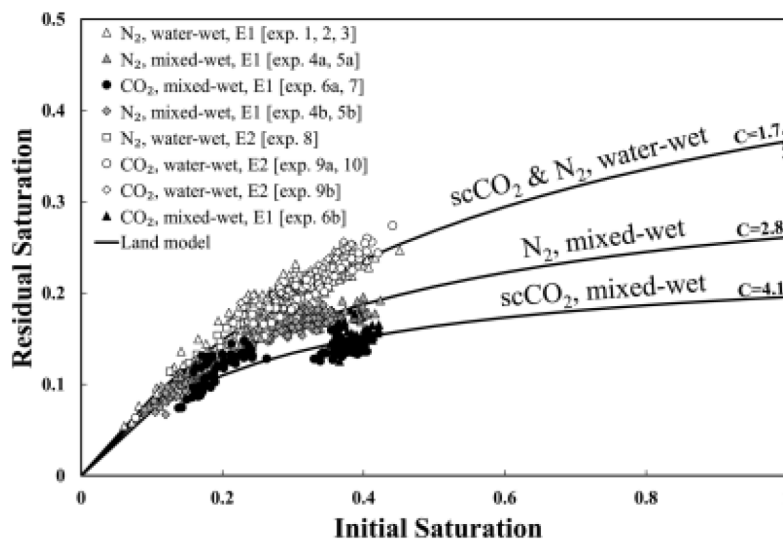


Figure 7—Initial and residual CO_2 and N_2 saturations measured. Solid lines show the best fit for experimental results using a land-trapping model (Al-Menhali and Krevor, 2016)

In addition to the wettability/contact angle and the IFT, the pore structure and porosity of the rock is also a critical consideration. Fig. 8 is produced by Iglauer et al. (2009) where different cores from various formations, rock types and porosities were tested for the maximum residual saturation and showed that the trapped saturation reduces as a function of increasing porosity. At low porosity, the rock has high P_c and hence a higher snap-off, yielding better residual saturation. So, the results presented here make intuitive sense that pore structure has a greater impact on CO_2 storage compared to the wettability.

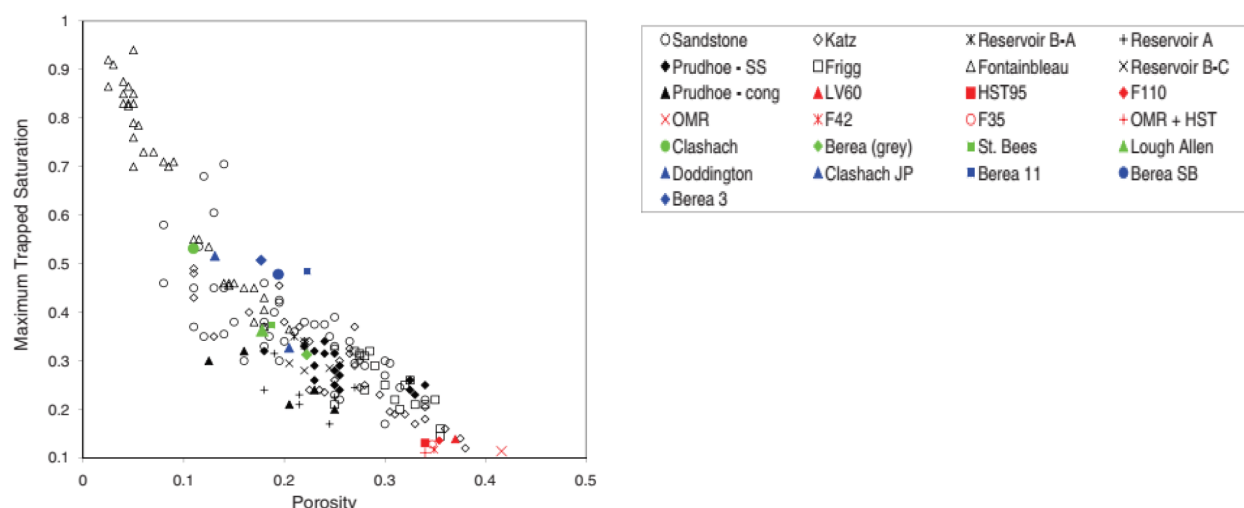


Figure 8—Maximum residual nonwetting-phase saturation as a function of porosity. (Iglauer et al., 2009)

For the consideration of potential storage after CO₂ fracturing, this result is extremely important. Higher porosity is a consequence of fractures, vugs and other similar features and hence this study tells us that the residual trapping potential will be lower. There are still benefits we can draw from the findings of this study. Firstly, the most crucial aspect in fracturing is that at larger reservoir scale, the after the fracture closes, the fluid migrates into pore structure that is far away from the fracture and could still be residually trapped there. Secondly, unconventional rocks have a very low porosity and become very viable candidates for CO₂ storage. For conventional rocks, the fractures will present a higher porosity but when the CO₂ leaks off and fracture closes, the structural trapping will be very high due to multiple barriers that exist above the fractured interval. Structural trapping hence becomes the dominant storage mechanism. Thirdly, when porosity is higher, the storage capacity will be higher, and a low residual trapping (% wise) might still mean high net equivalent CO₂ stored due to higher available volumetric capacity

Another critical question is, whether CO₂ alters the rock wettability when it is injected in the rock? Fig. 9 below is produced by Baban et al. (2021) from their NMR measurements and micro CT scans on quartz rich sandstone after the drainage and imbibition routine core flood (dead brine, live brine, scCO₂, live brine). It can be observed that the entire T₂ drainage curve (red) shifted towards left which indicates that CO₂ displaced the water from larger pores during drainage. This behavior is indicative of the rock remaining water wet because CO₂ is non-polar compared to water and hence it is difficult for CO₂ to displace water from the rock surface leaving a water film around the pores. But interestingly, the initially strongly hydrophilic sample turned significantly less hydrophilic when exposed to CO₂ either in form of molecularly dissolved or free scCO₂. Hence, CO₂ significantly reduces sandstone hydrophilicity and water-wettability. This implies that CO₂ will be imbibed deeper into the smaller pore throats and fracturing fluid filtrate will be recovered first in the fracture cleanup and flowback.

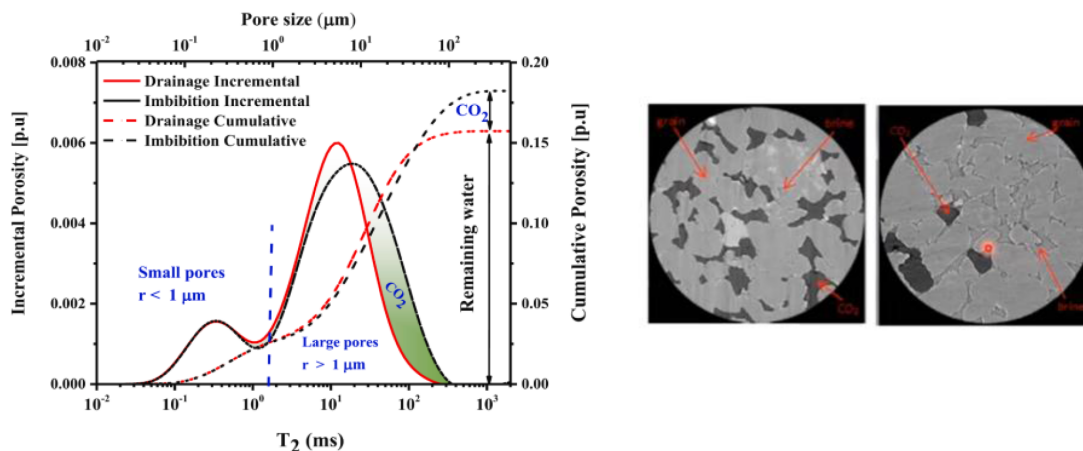
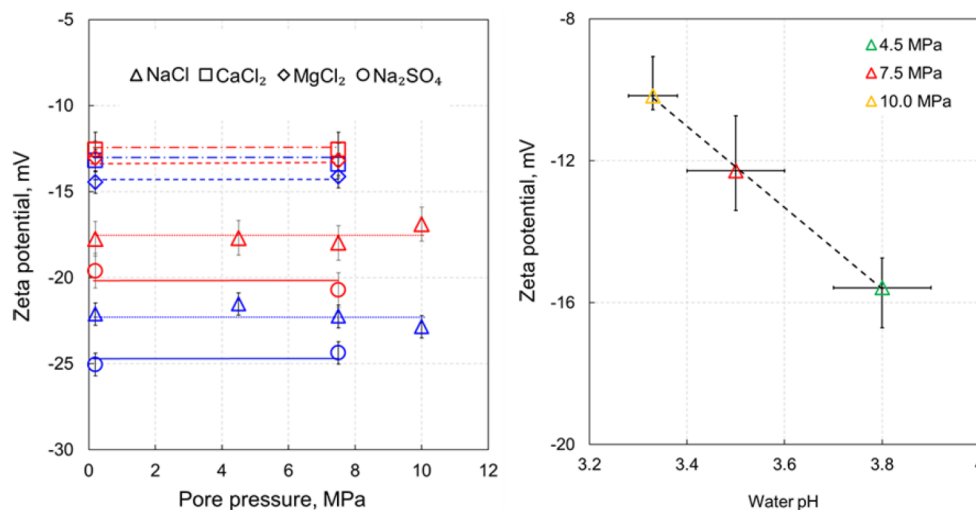


Figure 9—CO₂ core flood study results: (left) Water saturation in pore network of the San Saba sandstone after drainage (CO₂ flooding) and imbibition (secondary live brine flooding). (middle) CT scan result showing the initial CO₂ saturation. (right) CT scan result showing the residual CO₂ saturation. (Baban et al., 2021)

In addition to the core studies, molecular dynamics studies provide further insights into the wettability alteration after CO₂ injection. We understand that CO₂ is non-polar, but it does have polar interaction with water at the interface yielding a lower IFT and high solubility into water. Therefore, when CO₂ is injected in water, it will be dissolved, and the water is always in contact with the rock. Hidayat et al. (2022) have studied this interaction based on the streaming potential measurements at reservoir storage conditions on clean sandstone to measure the zeta potential, which is indicative of the evolution of rock surface charge. The work here started with injecting dead brine for a system pH of 7, at different pressures, different brine-types and the zeta potential was constant (Fig. 10, top left). Then live brine was injected (carbonated water) where the pH of the system is around 3.2 – 3.8. Here the pressure variation showed a variation in zeta potential because CO₂ solubility increases with the pressure (Fig. 10, top right). So, more dissolved CO₂ yields a lower pH and impact the zeta values.



The results in green experimental points in Fig. 10, bottom are representative of CO₂ storage condition because the pH expected due to CO₂ dissolution is in the range of 3 – 3.5 and shows that the wettability will be changed as a consequence. One limitation of this study is that only low salinity water was used because streaming potential depends on double layer which cannot be established in high salinity environment. This conclusion was further validated by Ali et al. (2022) with a similar setup by injecting different acid

concentrations. The authors found changing zeta potential values confirming the findings of Hidayat et al. that changing acidity of the rock as a consequence of CO₂ injection has the potential to alter the rock wettability.

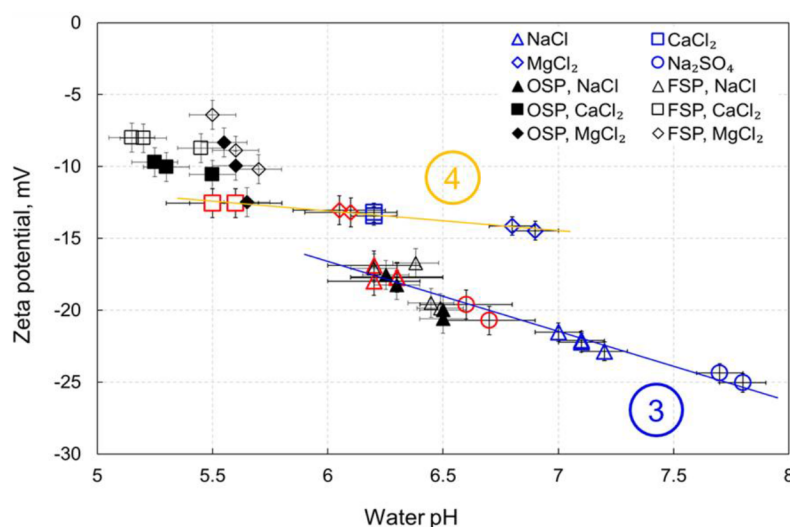


Figure 10—Surface charge impact of CO₂ injection: (top left) results for dead brine with a pH of 6-7. (top right) results for live brine for zeta potential changing with pH and pore pressure. (bottom) Zeta potentials measured versus pH of dead and live aqueous solutions. Different colors show the variation of pore pressures, temperatures, and salt types. The blue and red symbols correspond to 23°C and 40°C respectively. The green points are tests with carbonated water which is representative of the reservoir condition where CO₂ is being injected. (Hidayat et al., 2022)

Furthermore, to validate that the zeta potential and surface charge has a direct impact on the rock wettability and contact angle, Iglauer and Al-Yaseri (2021) conducted a study in basalt rock. An initial high contact angle rock was treated with a surfactant with a negative charge to increase the negative charge from -19 (original) to -36 (post-surfactant treatment) and Fig. 11 shows that the contact angle post treatment dropped down to 0 confirming that wettability can be impacted by CO₂ injection. An important consideration here is that in supercritical condition, the pH does not change considerably (Haghi et al., 2017), and the wettability alteration will be observed initially but then it will get constant after some time.

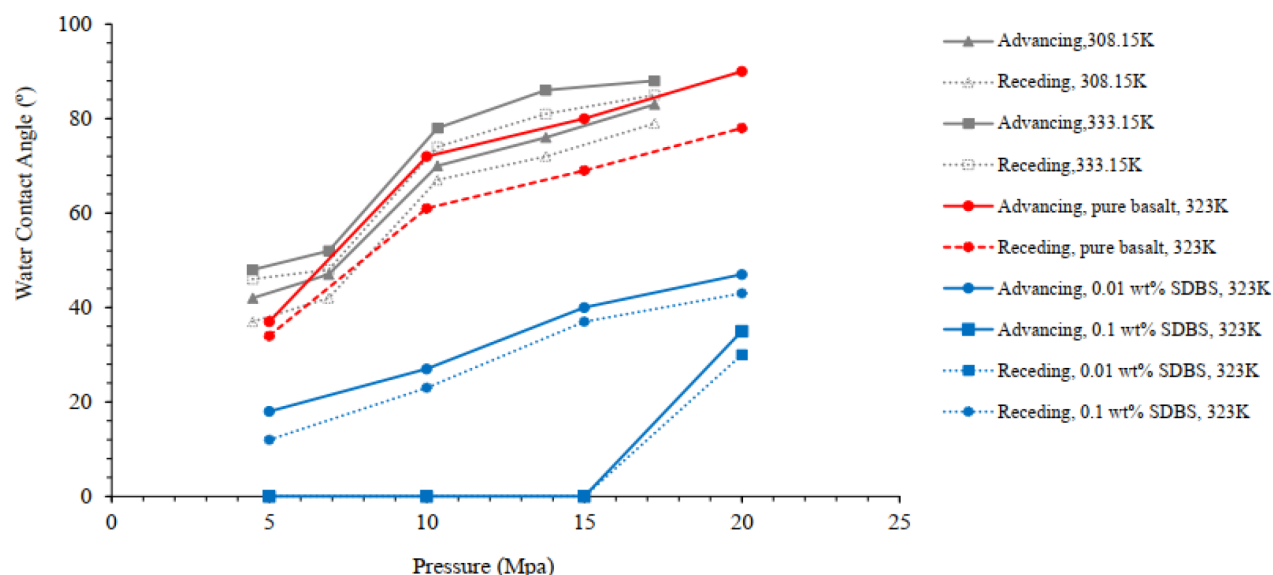


Figure 11—Water contact angles measured on the pure and surfactant-aged (SDBS) basalt samples in CO₂ atmosphere.

It is evident from all the literature that the CO₂ injection sites are all encountering lower pH and weak formic acid generation. This implies that all ongoing CO₂ effort in carbonate host rock will encounter rock wormholing due to the acid. As discussed above though that when highly porous features such as fractures, vugs, wormholes exist, residual trapping becomes less dominant and hence it would be no different for potential CO₂ storage after fracturing and stimulation. The advantage in this case though i.e., utilizing foam instead of pure CO₂, the pH will not go below 5 and hence a stronger cap rock integrity can be expected. Sealing efficiency will be better compared to a reservoir at an environment pH of 3 - 3.5.

Residual Trapping Studies

Fig. 12, left is taken from Burnside and Naylor (2014) and Fig. 12, right is taken from Zhang et al. (2017b) both the plots are showing the drainage and imbibition curves and they show the residual saturation of CO₂. This is the saturation at which the relative permeability of CO₂ to flow will be 0. This implies that, that fraction of CO₂ will be stored in the reservoir. The left figure shows that this saturation could be 48% but the authors publishing this data did extensive tests across 53 samples and showed that it could be as low as 5%. The low value is also supported in right figure where the residual CO₂ is 8-10%. This variation exists and depends on the heterogeneity of the rock sample itself. As explained earlier, the stored/residual CO₂ varies as a function of porosity, pore throat radius, permeability, pore structure, existence of vugs & fractures, rock wetting phase, fluid saturations before CO₂ is introduced, among other things.

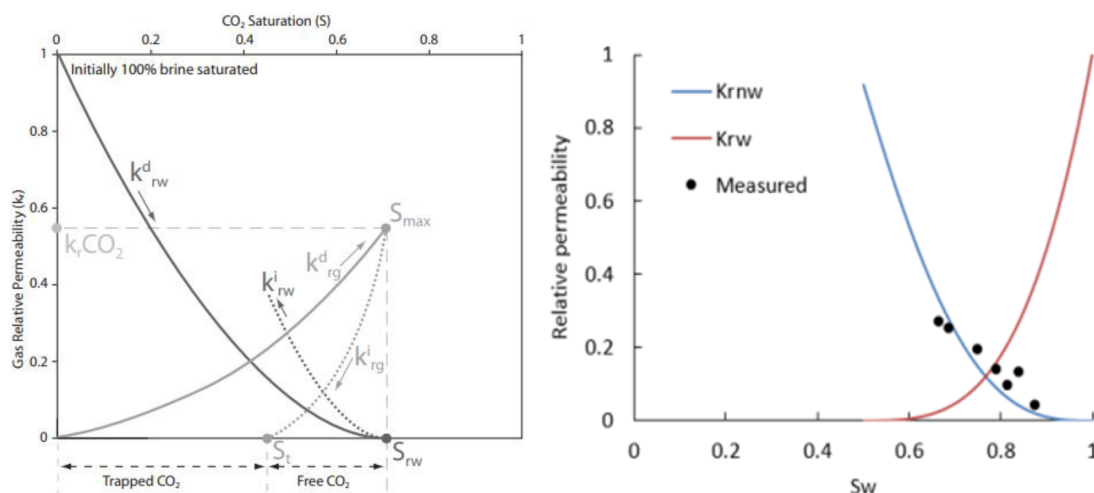


Figure 12—Core flood imbibition and drainage curves: (left) (Burnside and Naylor, 2014). (right) (Zhang et al., 2017b)

Niu et al. (2017) conducted a study with multiple core tests on Berea sandstone to study the effect of pressure, temperature and salinity on the initial versus residual trapping of CO₂ and found that there is no significant change in the CO₂ residual trapping. One limitation of this study is failure to establish an equilibrium of water with rock during the core flood and hence the change in saturation will not be expected without the equilibrium establishment. The equilibrium allows to capture the behavior precisely because the water is in a state of equilibrium with the reservoir rock.

If we scrutinize Fig.13, which shows the data when the well is put in production mode immediately after fracturing, i.e., the fracture cleanup or fracturing potential evaluation phase. The plot shows the evolution of CO₂ production, and it can be seen that in less than 4 days the CO₂ drops to ~0. Based on data analysis of more than 25 CO₂ treatments detailed by Almutary et al. (2021), the post-fracturing CO₂ return ranges between 5-20% of the total pumped amounts during the fracturing treatment. This analysis does not imply that 80-95% of CO₂ has been stored, and certainly higher fractions of CO₂ are produced in the lifecycle of the well. But based on the irreducible saturation of CO₂ during dynamic saturation changes in the production

phase, 100% of pumped CO₂ will not be produced. This data supports the physical concept of residual trapping of CO₂.

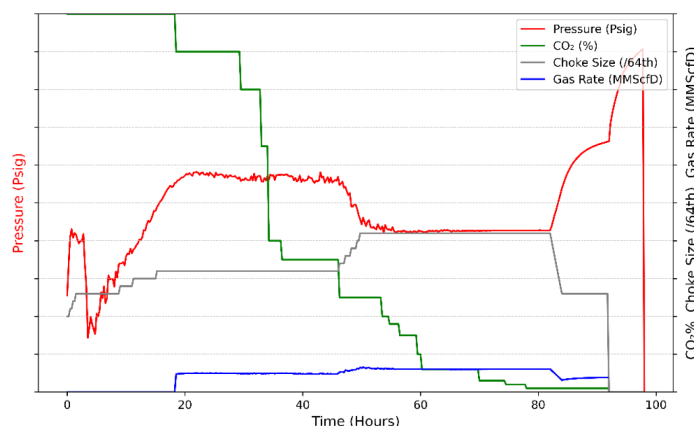


Figure 13—Flowback schematic for a typical CO₂ foam fracturing treatment showing the drop of CO₂ returns to 0 within first 3 days of fracture cleanup.

Enhancing the Residual Trapping

This section discusses a research and engineering avenue for increasing the stored/trapped CO₂ through residual trapping by shifting the drainage curve with the application of chemistry to modify relative permeability of CO₂. Use of relative permeability modifiers (RPM) is common in the oil and gas industry but there has never been an RPM utilization approach with CO₂ assisted fracturing or intervention for the purpose of reducing CO₂ recovery post-treatment. Fig. 14 shows the conceptual diagram for this approach where the residual CO₂ saturation shifts, reducing the recovery of CO₂ significantly and eventually storing more CO₂ pumped in fracturing, stimulation, and intervention treatments. The engineering workflow involves testing the RPM additive for compatibility with reservoir fluids and with fracturing, stimulation, intervention fluids to ensure it does degrade or negatively affect the quality of these base fluids. The RPM additive can also be pumped at periodic intervals of the production lifecycle of the well to ensure the CO₂ is stored permanently.

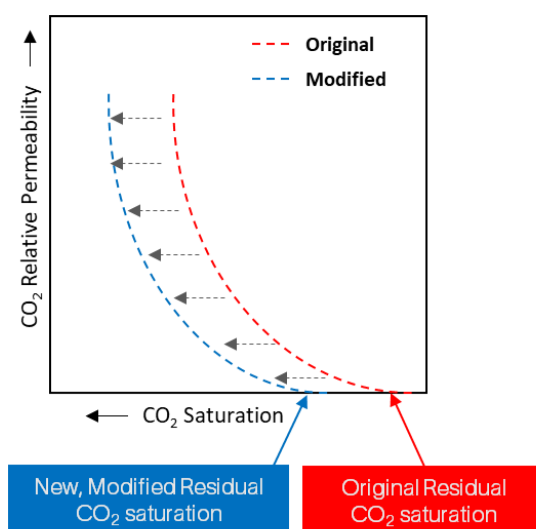


Figure 14—Conceptual schematic of a CO₂ relative permeability modifier.

The primary challenge here is no such commercial chemistry solution exists today to implement an RPM for CO₂ and utilize it as an additive along with fracturing, stimulation, and intervention fluid systems. But some concepts have been explored which include absorbent agents which chemically interact with CO₂ to increase residual trapping, and various surfactants have been developed in CO₂ EOR projects which aim to reduce the capillary forces by 4 orders of magnitude from 10 mN/m down to 0.0001 mN/m (Chabert et al. 2016). This reduces the capillary number of water and minimizes its trapping retention hence enhancing CO₂ trapping.

The research and engineering around this development can be systematically addressed for potential concepts with an engineered workflow shown in Fig. 15. The figure shows the critical steps in the development-to-implementation process. The process includes conducting core tests, rheometer tests and fluid compatibility tests to ensure that (a.) CO₂ recovery can be reduced efficiently, (b.) the novel additive and the system it is mixed with, are fully compatible with the reservoir fluids. Compatibility here entails that undesirable byproducts such as emulsions, sludges are not generated, (c.) the novel additive is compatible with the fracturing, stimulation, or the intervention fluid system. Compatibility here entails that it does not degrade the fluid reducing its viscosity which would mean the fracturing (or any other end treatment objective) may be hampered. (d.) the regained permeability does not reduce a lot compared to the baseline.

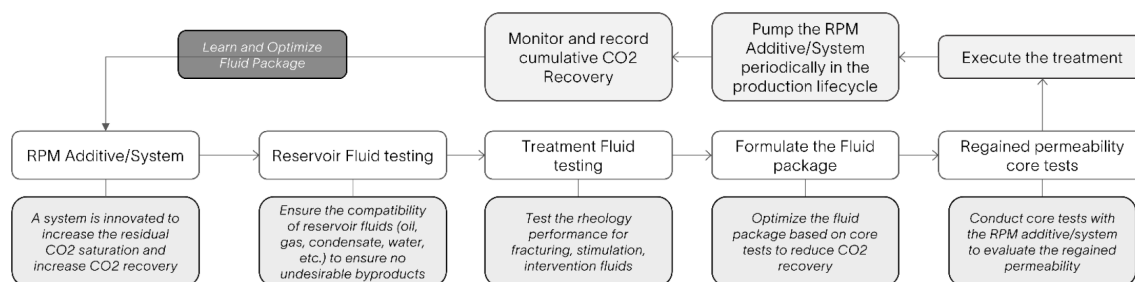


Figure 15—Research workflow for innovating, testing and validating the relative permeability modifier solution.

Once the testing and quality checks are conducted, treatment execution and following long term measurements and calibrations can be put in place to ensure that there is a continuous improvement loop. Also, with more experimental, execution and post-treatment data accumulated, a database should be maintained for possible statistical and machine learning based optimizations for the future.

There have been attempts to work on similar RPMs to enhance the CO₂ storage capacity and most of the research aims at making the reservoir more water wet but while doing that, the IFT could be reduced. This becomes counter intuitive because contact angle variations are affected by cosine function but the dependence of Pc on IFT is direct and hence will impact significantly. One such study was done by Al-Yaseri et al. (2021) where silica nanofluids were used as an additive to change the rock wettability to water wet. Their findings are summarized in Fig. 16 which illustrates this trade-off. While the SNP addition shifts the system toward more water-wet conditions (lower contact angles, favorable for residual trapping), it simultaneously reduces the CO₂–brine interfacial tension. Because capillary entry pressure scales directly with IFT, this reduction lowers the maximum CO₂ column height that can be supported beneath the caprock. A shorter column height implies a higher risk of buoyant CO₂ escape and hence a decrease in overall trapping security. Thus, wettability alteration toward water-wetness, if accompanied by IFT reduction, may counterintuitively diminish storage capacity.

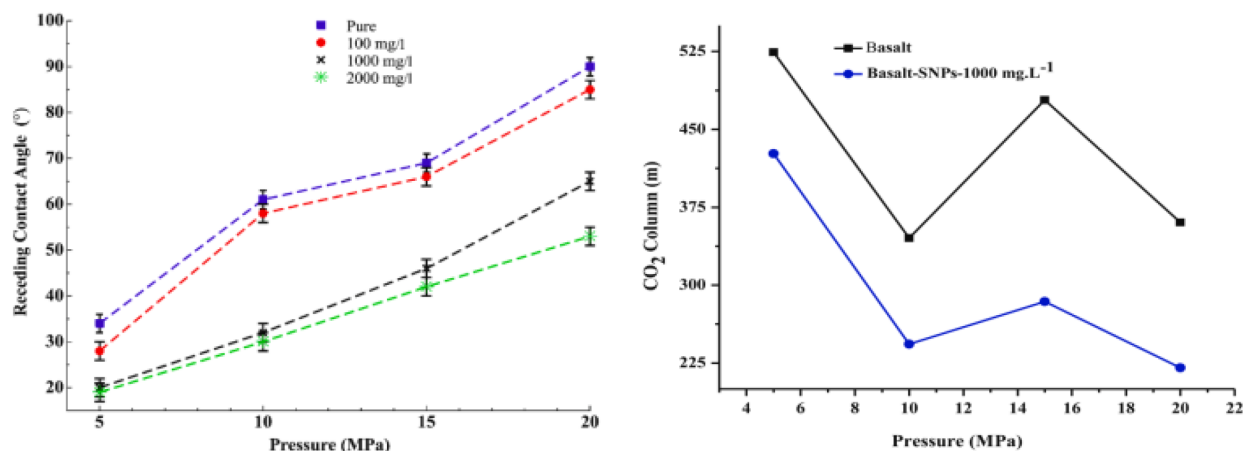


Figure 16—Consideration of the interdependency of IFT and contact angle: (left) Receding water contact angle for CO₂/water/basalt at 323K systems as a function of pressure and additive concentrations. (right) Maximum column height as a function of pressure. Black is before treatment and blue is after treatment. (Al-Yaseri et al. 2021)

Engineered Workflow for Implementation

The literature and research discussed includes statistical, laboratory experimentation, and field observations. The results demonstrate the potential of retaining show that a high fraction of CO₂ is retained but in the consideration for fracturing and stimulation, there are some variations. These variations include, (a) the impact on flow behavior of multiphase flow with fracturing fluid filtrate, scCO₂, reservoir fluids, etc. (b) the rock has features such as fractures or wormholes which could impact these results as well. Hence one of the challenges around this approach is investigating the ability to store CO₂ after fracturing or stimulation is done. This challenge can be systematically addressed with an engineered workflow shown in Fig. 17. This workflow can be used to model the CO₂ storage underground and the quantification process related to it. The figure shows that we start with (a.) core tests to study the fluid saturation and relative permeability behavior under CO₂ drainage and imbibition. The relative permeability for fracturing filtrate, CO₂ and reservoir fluids can be quantified in the core study. (b.) The results of the core experiments will be inputs into the numerical model for production flow simulation. The setup for the post-fracturing simulation can be populated with a dynamic fluid saturation simulation prepared upstream of this step. The model can be initialized with end of fracturing treatment saturations along with the fracture modeled (explicitly or numerically simulated) in the reservoir grid. This will conclude the design and preparation phase and will allow the design engineer to tailor/optimize the design. (c.) Next step is the wellsite execution of the treatment. (d) After the execution, post-treatment measurements are crucial for the pilot phase. Measurements can be made with multiphase flowmeters (Alharbi et al. 2022), venturi correlations or inline CO₂ sensors to measure and track cumulative CO₂ recovery/storage. (e.) Long term subsurface understanding is also crucial with measurements using a monitoring well, distributed sensing techniques to model long term CO₂ plumes. The post-fracturing and long term measurements data can be used to quantify CO₂ recovery/storage and fed back to the numerical engines for calibration. Similar impact quantification can also be done for the use of RPM additives. Once the static reservoir model is calibrated with real-data, the CO₂ storage estimates can be upscaled to the assets.

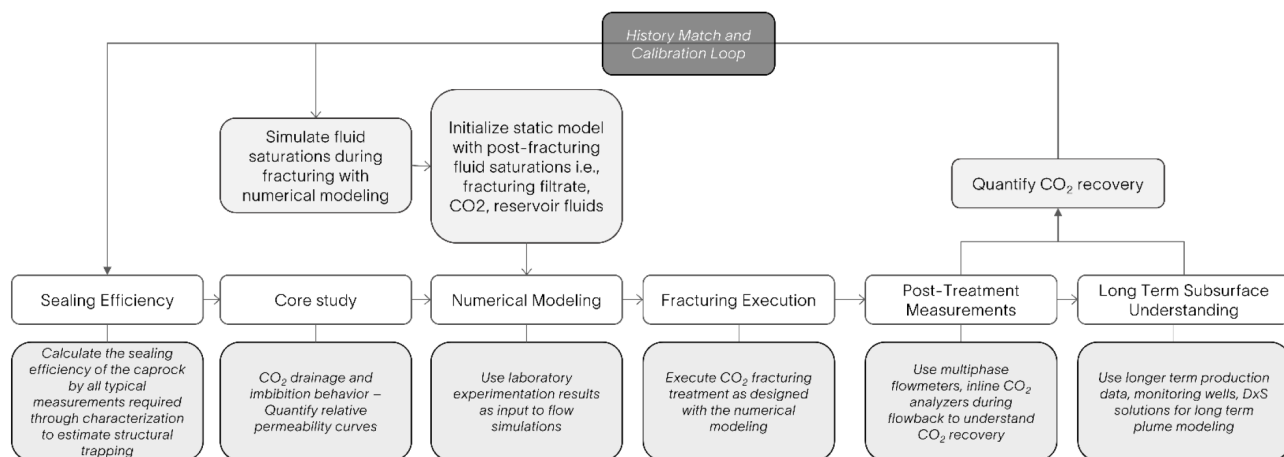


Figure 17—Engineered end-to-end workflow to evaluate and model CO₂ flow behavior and retention associated with CO₂ fracturing and stimulation.

Leveraging Wellsite Capturing Technologies

The objective of this concept exploration is to reduce the overall carbon footprint in an effective engineered way, but for CO₂ utilization in fracturing, complexity, time and cost components exists at each level of this value chain. The key issues are highlighted below –

1. CO₂ is captured from multiple industrial sources or directly from the air itself. This captured CO₂ requires compression and storage until it's time for utilization.
2. In the scenario where a continuous injection stream is established between the capturing and storage sites, it requires significant capital investment for the infrastructure because large-scale transport via pipeline and ship is required which also adds operating expenses for maintaining the infrastructure periodically. For smaller volumes, truck or rail is used which increases the cost per ton of CO₂. Moreover, all the transport options add incremental carbon emissions through fuel emissions and adds to the net carbon balance.
3. Storing CO₂ at an intermediary location before its use also requires compression and the energy consumption of compressors adds incremental carbon.

A new workflow/process can also be utilized to overcome these challenges mentioned above by addressing each component of the CCUS in an innovative way. The components of utilization and sequestration of CO₂ have been detailed in the previous sections. The remaining component of capturing CO₂ can be done from the wellsite equipment itself during pumping operations and transferred to a CO₂ storage tank which can then be moved immediately (after the job is done) to another wellsite located close, requiring hydraulic fracturing or intervention (Fig. 18). Optimization algorithms based on grid based search and path planning can be used such as Dijkstra's algorithm, A* search (with Euclidean distance as heuristic), concurrent variation of Dijkstra's algorithm, etc. to ensure the CO₂ is being moved to the most efficient distances (Narang et al. 2024).

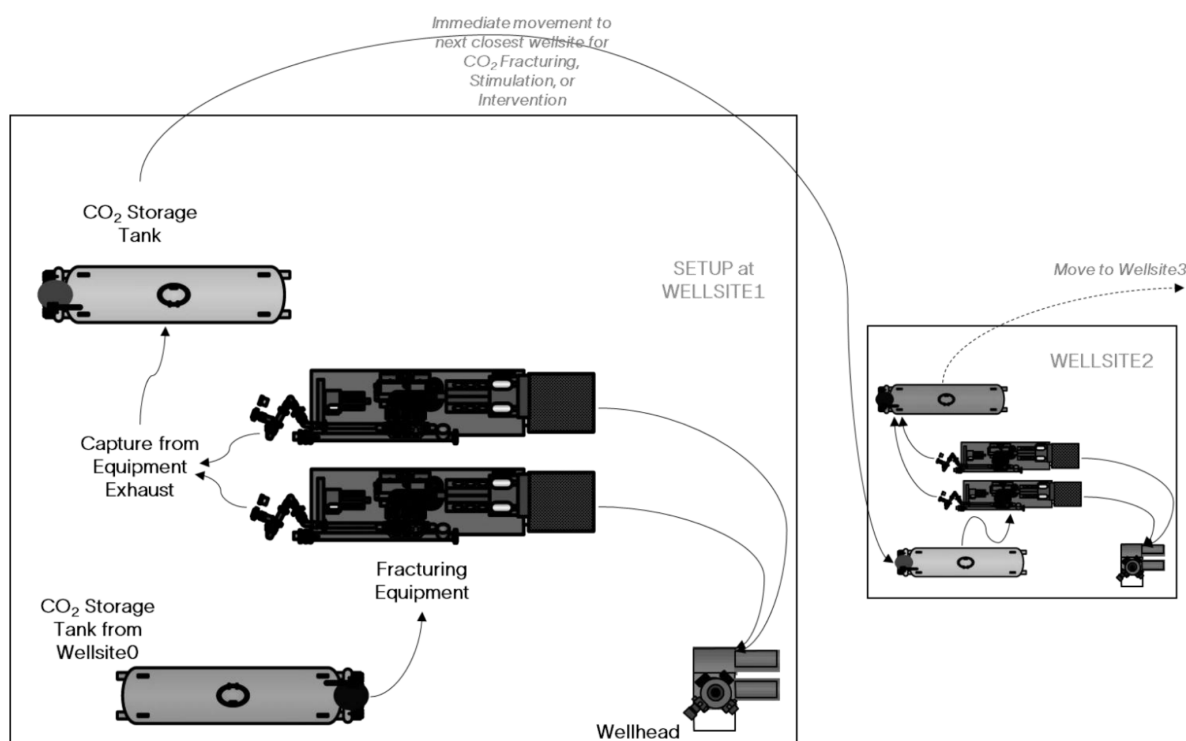


Figure 18—Lifecycle utilization for CO₂ capturing and fracturing to minimize the emissions associated with CO₂ transportation.

This solution integrates (a.) reduction of the lifecycle of carbon, (b.) reduction of the incremental carbon released per CO₂ sequestered, and (c.) reduction of the overall investment of typical CCUS projects with an alternate solution targeting the same principles and objectives –

1. The immediate movement from wellsite to wellsite within the same asset reduces the time of carbon on surface.
2. By minimizing the distance of CO₂ movement from capturing and pumping site, the transportation and logistics associated fuel emissions are reduced.
3. The emissions involved in compressing and storing CO₂ at intermediate sites before utilization can be significantly reduced.

Global Potential

The concept of CO₂ fracture sequestration has a significant global potential, given the widespread use of hydraulic fracturing and the vast number of suitable reservoirs worldwide. Unconventional oil and gas development has exploded in recent decades. Shale basins and tight formations exist on every continent (major basins in North America, China, Argentina, Australia, Middle East, etc.), representing thousands of fracturing candidate wells. If even a portion of new fractures or re-fracturing operations utilize CO₂, the cumulative CO₂ storage could be enormous. A recent assessment highlighted that if water-based fracturing were broadly replaced by supercritical CO₂ fracturing, a single horizontal well could store on the order of 10,000+ tons of CO₂ (Hou et al. 2024). Considering that large oil/gas fields often involve hundreds or thousands of fractured wells, the scalable storage capacity runs into the millions of tons per field (Hou et al. 2024). These numbers might be assuming a higher CO₂ fraction than practically possible for the available supply of CO₂ but still imprints the scale of CO₂ storage. Across the globe, where tens of thousands of wells are fractured each year, CO₂ fracturing could potentially sequester gigatons of CO₂ over time while maintaining energy supply. Compared to the claimed storage amounts from the Sleipner project which is 0.9

Mt of CO₂ per year (Sleipner fact sheet, 2016), this approach makes it a compelling strategy for countries looking to meet climate targets without halting oil and gas production immediately.

To put this in perspective, the International Energy Agency (IEA) estimates the world may need to be storing on the order of 5,000 Mt (5 Gt) of CO₂ per year by 2050 to meet climate goals (Malischek and McCulloch, 2021). Harnessing existing oil and gas operations for a portion of this via CO₂ fracturing could contribute significantly. Many of the world's largest CO₂ emitters (e.g. China, U.S., India) also have large unconventional hydrocarbon resources (Sovacool et al. 2014). China has extensive shale formations (like Sichuan, Ordos basins) and is actively researching CO₂ fracturing as a way to both stimulate difficult reservoirs and store CO₂ from coal-fired plants. Pilot projects in China's Jilin oilfield and elsewhere have already demonstrated successful CO₂ fracturing with enhanced oil recovery and storage (Tao et al. 2021), and more are underway. In North America, the huge scale of shale operations (Permian, Marcellus, Eagle Ford, etc.) presents a ready opportunity. If tax incentives or carbon prices support CO₂ usage, even a fraction of wells doing CO₂ fracs could sequester tens of millions of tons annually while yielding continuous hydrocarbon production. Other countries with emerging shale industries (Argentina's Vaca Muerta, Australia's basins, Middle East tight gas, etc.) could leapfrog to low-water, low-carbon fracturing solutions from the start.

Realizing these numbers and scale globally will certainly depend on overcoming technical and economic challenges. Issues like CO₂ supply logistics, cost of CO₂ vs. water, and ensuring effective proppant transport at large scale must be managed (Hou et al. 2024). There are also safety considerations (CO₂ at high pressure is cryogenic and asphyxiant if leaked). However, trends are encouraging: as carbon prices rise and emitters seek disposal options, more CO₂ will be available for use. Technology around fracturing fluids to keep stable CO₂ foams is also improving. Notably, researchers propose strategies like blending CO₂ with traditional fluids or pre-injecting CO₂ before water (CO₂ pre-pad) to gain partial benefits in a transition phase (Li et al. 2022). Such hybrid approaches could ease CO₂ fracturing into mainstream use. Additionally, this cool pre-pad injection also causes a much faster near wellbore cooldown which can enable fluid stability for higher exposure times in extremely high temperature reservoirs.

If these advances continue, CO₂ fracturing could become standard practice in many regions, transforming tight gas and unconventional reservoirs into sizable carbon sinks. The vision is that hydrocarbon production and recovery enhancement go hand-in-hand with carbon storage. Instead of fracturing with water and emitting CO₂, operators will inject captured CO₂ to fracture, leave a portion sequestered, and produce cleaner energy. Scaling this will directly contribute to climate mitigation. In summary, the industry needs to leverage the extensive footprint of oil and gas development to deploy CCUS on a distributed, incremental basis. Every shale basin or tight oil field can be seen as an opportunity to store carbon. It aligns especially well with countries that have both climate goals and abundant tight gas and unconventional reserves, offering a pathway to sustainable hydrocarbon use. With further field experience and supportive policy through carbon credits for stored CO₂, this technique could revamp the current energy scenario, turning a fracturing often seen as environmentally challenging into a tool for carbon management on a global scale.

CO₂ Emissions Reduction

A core question here is how does the total emissions footprint compare for conventional fracturing versus CO₂ foam fracturing? The answer has multiple components – direct sequestration of CO₂, avoided emissions from water hauling, and reduced flaring due to faster flowback of fracturing fluid filtrate and fracture cleanup. The author's upcoming work will be on a detailed lifecycle assessment for CO₂ and N₂ foam fracturing versus conventional fracturing.

Direct CO₂ Sequestration

Each CO₂ frac job has the potential to sequester a significant mass of CO₂ in the subsurface. Let us consider a horizontal shale well fracturing operation using ~15000 tonnes of CO₂ (rough calculations considering 10,000-ft lateral × 2,000 bbl/ft fluid intensity × 20% CO₂ considering supply chain issues). If a conservative 25% of that CO₂ is retained (Bello et al. 2024), that's ~3,750 tonnes of CO₂ permanently stored per well. Bello et al. (2024) demonstrates achieving ~40% CO₂ trapping based on foam injection studies. Here the assumption of 25% is justified due to changing transport mechanisms in a fractured reservoir versus intact matrix. To put that in perspective, 3,750 tonnes CO₂ is roughly equivalent to the annual emissions of 800 cars or the CO₂ to generate electricity for 560 homes for a year. Scale that by hundreds or thousands of wells, and the numbers reach millions of tonnes stored – comparable to large point-source CCS projects. Moreover, with appropriate research towards a reliable CO₂ relative permeability modifier, the 25% can scale up to 50% or more CO₂ retention in subsurface. Even smaller frac jobs (e.g. a refrac on a conventional well using 500 tonnes CO₂) could store > 100 tonnes CO₂ each, still a noteworthy offset. Over an entire field development of dozens of wells, the cumulative sequestration could offset a significant portion of the field's lifecycle emissions. In fact, analyses of CO₂ EOR operations found that if done efficiently, the net carbon footprint of the produced oil can be near-neutral or even negative, i.e., storing more CO₂ than the oil's end-use emissions (Bello et al. 2024). As an analogue, CO₂ fracturing, which tends to store even more CO₂ proportionally than CO₂ EOR could yield net-zero oil or gas production from those wells. This means the CO₂ injected and left in the ground compensates for the emissions during fracturing as well as the production and consumption processes related to the produced hydrocarbon.

Avoided Emissions and Environmental Impacts

Using CO₂ in place of water can also indirectly reduce emissions and environmental harm. Water management in fracturing has a carbon footprint. Energy is used to pump and transport millions of gallons, and to treat or dispose of flowback water (Khan et al. 2025). By eliminating most of the water, we avoid the truck traffic and pumping energy associated with it. As a replacement, the CO₂ transport emissions are much lower due to less transport trips required. That translates to lower diesel combustion and thus CO₂ emissions. Additionally, the avoidance of water disposal sidesteps the induced seismicity and contamination risks tied to those operations. These co-benefits, while harder to quantify in CO₂ terms, contribute to an overall more sustainable profile for the fracturing operation. Moreover, if CO₂ fracturing allows production in water-scarce regions that otherwise would require energy-intensive water sourcing (e.g., trucking water from far away or desalinating water), the net energy and emissions balance improves.

Economic Considerations

This approach has multiple economic considerations and financial implications that demand separate research and analysis. It has multiple benefits and tradeoffs that need to be evaluated in detail for specific applications. Below are a few points for discussion -

Overall Asset Utilization

Enhanced Overall Hydrocarbon Recovery: Depleted reservoirs which have reached their hydrocarbon recovery factor are commonly selected for CO₂ storage due to their proven seal integrity, existing infrastructure, and well-characterized geology. When CO₂ foam is introduced as a stimulation fluid, the operator can extend the life of the asset by enhancing the recovery factor of the remaining hydrocarbons in place. Foam improves CO₂ sweep efficiency by reducing gas mobility and mitigating channeling, thereby contacting previously bypassed oil or gas zones. This not only prolongs the economic life of mature fields but also generates incremental revenues during the transition to storage operations.

Transition to Dedicated Storage: Once the economically recoverable hydrocarbons are produced, the same fracture network created during CO₂ foam fracturing serves as an enhanced storage volume. Fractures increase the accessible pore volume and effective injectivity of the formation, raising the initial CO₂ storage capacity compared to the original unstimulated, depleted reservoir. This dual function converts the asset into a more reliable long-term CO₂ sink without requiring major new capital investments in subsurface development.

Hence, for operators, it is an economic "win-win" Scenario. This integration creates a stacked revenue model with, (a) upstream cashflow from incremental oil and gas recovery via CO₂ foam-stimulation, (b) carbon credits or tax incentives for CO₂ storage (e.g., U.S. 45Q credits, EU ETS allowances), (c) extended asset life by repurposing existing wells and facilities, reducing abandonment liabilities.

Economic analyses of CO₂-EOR already show that with supportive carbon pricing, projects can be carbon-neutral or carbon-negative (Azzolina et al., 2015). The addition of CO₂ foam stimulation enhances the business case by simultaneously improving recovery and retention efficiency (Al-Menhali and Krevor, 2016). Thus, the approach offers asset owners a risk-hedged pathway to both monetize hydrocarbons and participate in decarbonization markets. Moreover, operators adopting this approach may benefit their strategic positioning from reputational and ESG (Environmental, Social, and Governance) advantages. By framing mature field operations as part of a carbon management strategy, companies can attract investment aligned with low-carbon portfolios while still generating value from existing assets.

Opportunity for Tax Credits

Under U.S. tax law, Section 45Q provides credits for qualified CO₂ utilization and storage activities. Specifically, IRC Section 193(b)(3) defines "tertiary recovery methods" to include techniques designed to increase hydrocarbon recovery beyond primary and secondary stages, such as CO₂ flooding, miscible injection, and thermal methods. Currently, the framework explicitly recognizes CO₂ EOR as an eligible utilization pathway. However, with the effort presented in this paper of compiling research references, the CO₂ foam fracturing presents a compelling analog. It shares many similarities such as operation in pressure-depleted zones, enhancement of recovery factors, and net storage of CO₂, but also provides unique benefits, including new fracture creation, enhanced permeability and storage capacity, and reduced flowback volumes compared to conventional EOR.

To secure eligibility for 45Q credits, operators and industry consortia would need to approach regulators with technical evidence demonstrating that CO₂ foam fracturing (and by extension refracturing in multiple decline unconventional shale assets) qualifies as a tertiary recovery method under the statutory definition. The case is strengthened by the fact that, like EOR, this technique results in incremental oil/gas recovery while permanently sequestering a portion of injected CO₂. The engineering roadmap laid out in this paper covering fluid design, residual trapping validation, and field demonstration forms the backbone of the proof-of-concept required by the IRS and Treasury.

Similarly, in the European Union, the Emissions Trading System (EU ETS) allocates carbon credits for verified CO₂ storage projects, with CCS in saline aquifers and depleted fields already recognized. For CO₂ foam fracturing, the regulatory case is analogous, i.e., the storage is permanent, quantifiable, and verifiable, and the hydrocarbon recovery function does not undermine its qualification. In EU, since fracturing has an additional layer of regulatory requirements, this approach will still be valid to matrix acidizing and acid fracturing with foams, which is recurrent operations in EU, offshore. One advantage in the EU ETS framework is the growing emphasis on "CCU" (carbon capture and utilization). If operators can show that stimulation leads to both utilization (hydrocarbon recovery) and durable storage, it should be possible to classify the technique alongside.

To formalize CO₂ foam refracturing as an eligible pathway, industry stakeholders should:

- Submit pilot project data quantifying incremental recovery and net storage.
- Demonstrate monitoring frameworks (e.g., CO₂ flowback quantification, tracer studies, pressure-volume balance) to regulators.
- Engage in rulemaking petitions to the IRS (U.S.) and European Commission (EU) to expand existing definitions of tertiary recovery or utilization/storage.
- Leverage analogies to EOR by emphasizing operational similarities while documenting the additional residual trapping and storage potential unique to refracturing.

Summary and Conclusions

A typical application of effective utilization would be the use of CO₂ in an enhanced oil recovery. But another application, i.e., fracturing and intervention jobs also require treatment fluids and CO₂ could be utilized in this by replacing a fraction of fracturing, stimulation or intervention fluids. Utilization principle for stimulation is extensively studied and established because it enhances production performance, speeds up the fluid recovery, and reduces freshwater utilization. In fracturing applications, multiple cap rock layers exist, which, with high capillary forces make for robust structural trapping for the CO₂ pumped during fracturing, stimulation or intervention. Also, residual trapping potential due to CO₂ residual saturation and reduced relative permeability to flow exists. This supports the idea of not recovering all the CO₂ on surface during the production lifecycle of the well.

The exploration of CO₂ foam fracturing as presented reveals a new paradigm where hydraulic fracturing can be hypothesized (with compelling evidence) as a carbon utilization and storage process rather than purely carbon intensive. In this hypothesis, by replacing water with CO₂, we can simultaneously enhance hydrocarbon production and sequester substantial CO₂ in the subsurface, and the evidence in the literature along with the research and development roadmaps strongly supports this vision. CO₂ as a fracturing fluid offers unique production advantages that have been demonstrated in field trials ([Almutary et al. 2021](#); [Tao et al. 2021](#)), addressing the utilization objective. At the same time, the injected CO₂ can be trapped by multiple mechanisms in the reservoir (structural, residual, solubility, etc.), with engineered approaches like foam significantly boosting the fraction retained from ~20% in traditional CO₂ EOR up to ~40–50% in foam-assisted scenarios ([Bello et al. 2024](#)). If realized, this implies each CO₂ fracture and stimulation candidate could become a net sink for carbon even as it yields oil or gas.

The proposed paper structure delineated the key elements needed for a coherent academic argument: starting from global CCUS context and moving through detailed technical discussion of CO₂ fracturing benefits, sequestration mechanics, practical implementation, and global scaling implications. Each section contributes a piece of the narrative that demonstrates feasibility for the hypothesis addressing the energy supply and emissions reduction challenges. Multiple enabling factors already exist such as suitable reservoirs, large CO₂ sources, field equipment for CO₂ fracturing, etc. Moreover, fracturing itself inherently solves many storage site screening criteria. The engineered workflow provided a roadmap for testing and deploying this method, ensuring that it can be validated rigorously (with modeling and monitoring) just like any CCUS project.

Typical CCUS projects involve a lot of capital investment for design, execution and measurements involved in well construction, and the digital and physical tools required for evaluating caprock integrity, modeling & calibrating CO₂ plume, which can all be eliminated with this approach. Additionally, the benefit of this technique ensures the storage requirements better than typical sites which tend to be shallow sandstone aquifers. The performance is better in terms of -

- a. Stronger cap rock integrity because multiple barriers exist with high capillary forces.
- b. Better storage capacity due to enhanced fracture area/volume

- c. No requirement to test the injectivity because that is already checked for during fracturing, stimulation, and intervention.
- d. Well integrity issues.

Certainly, technical challenges exist, for instance, achieving fluid viscosity with foams at high temperatures ensuring adequate proppant transport, controlling fracture growth, developing cost-effective mobile CO₂ capture for pump emissions, and sourcing sufficient CO₂ infrastructure. These are non-trivial gaps, as identified by Hao et al. (2014). However, none appear unachievable given current R&D implementations (Khan et al. 2023a, etc.). Novel thickening agents for CO₂ and improved foam surfactants are under active study, and the oil industry has decades of experience handling high-pressure CO₂. Economic considerations will also dictate the pace of adoption; the approach will be most attractive where carbon prices or ESG drivers reward CO₂ storage, or where water constraints force a re-think of fluid choice.

In conclusion, CO₂ foam fracturing stands out as a strategic convergence of petroleum engineering and carbon management. It has the potential to transform conventionally high-footprint operations into climate-friendly, effectively using the subsurface as a giant carbon sink at the very moment we unlock hydrocarbons flow from it. This dual outcome encompassing enhanced production and emissions mitigation makes it a particularly elegant solution as society navigates the energy transition. Moving forward, field pilots implementing the ideas from this "engineered workflow" will be critical to demonstrate real-world feasibility. If those pilots mirror the successes hinted in literature, CO₂ fracturing could indeed change the fracturing market and narrative.

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